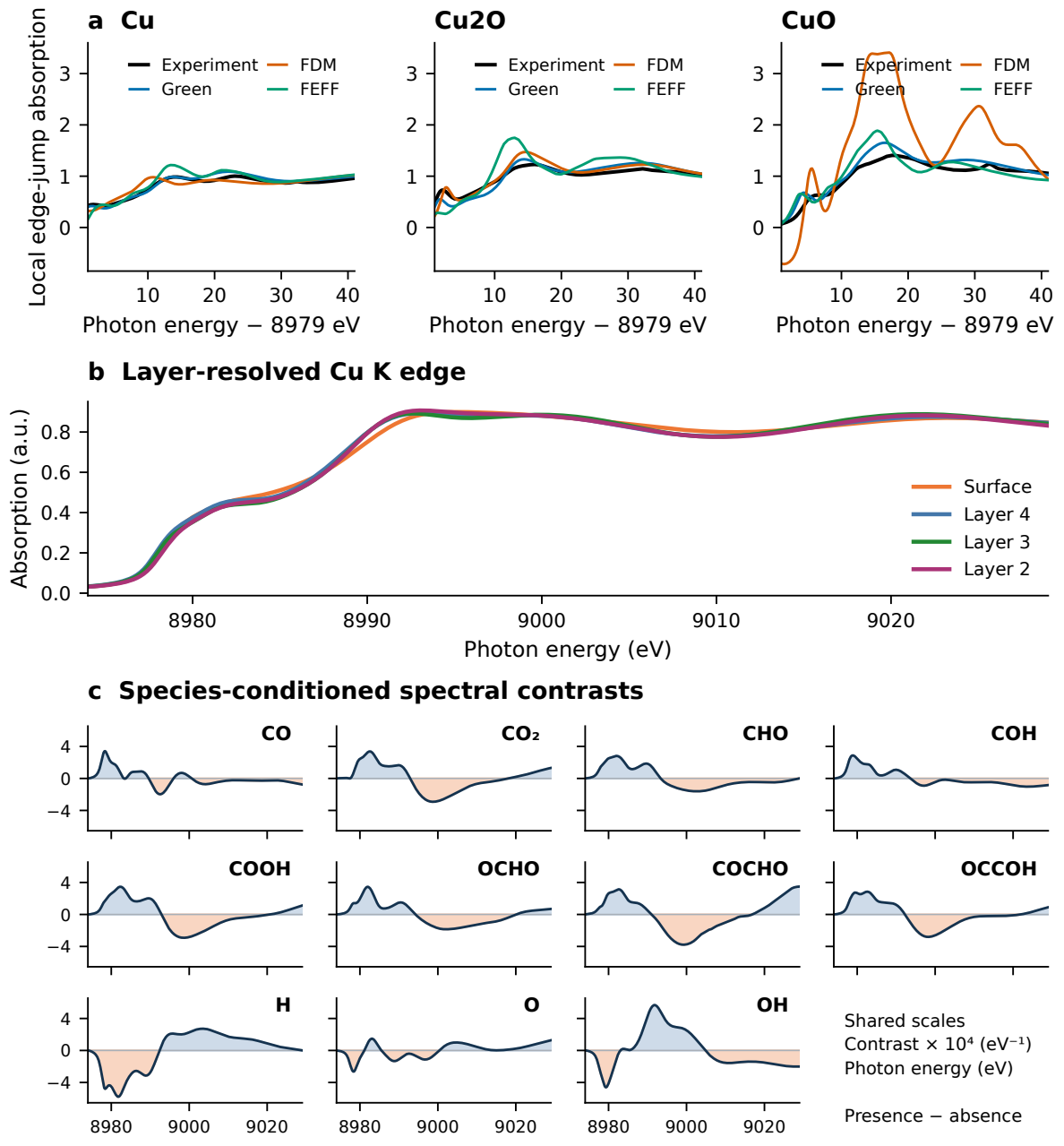

a Cu
Cu2O
CuO
Experiment
FDM
Green
FEFF
Local edge-jump absorption
Photon energy − 8979 eV
b Layer-resolved Cu K edge
Absorption (a.u.)
Surface
Layer 4
Layer 3
Layer 2
Photon energy (eV)
c Species-conditioned spectral contrasts
CO
CO₂
CHO
COH
COOH
OCHO
COCHO
OCCOH
H
O
OH
Shared scales
Contrast × 10⁴ (eV⁻¹)
Photon energy (eV)
Presence − absence